

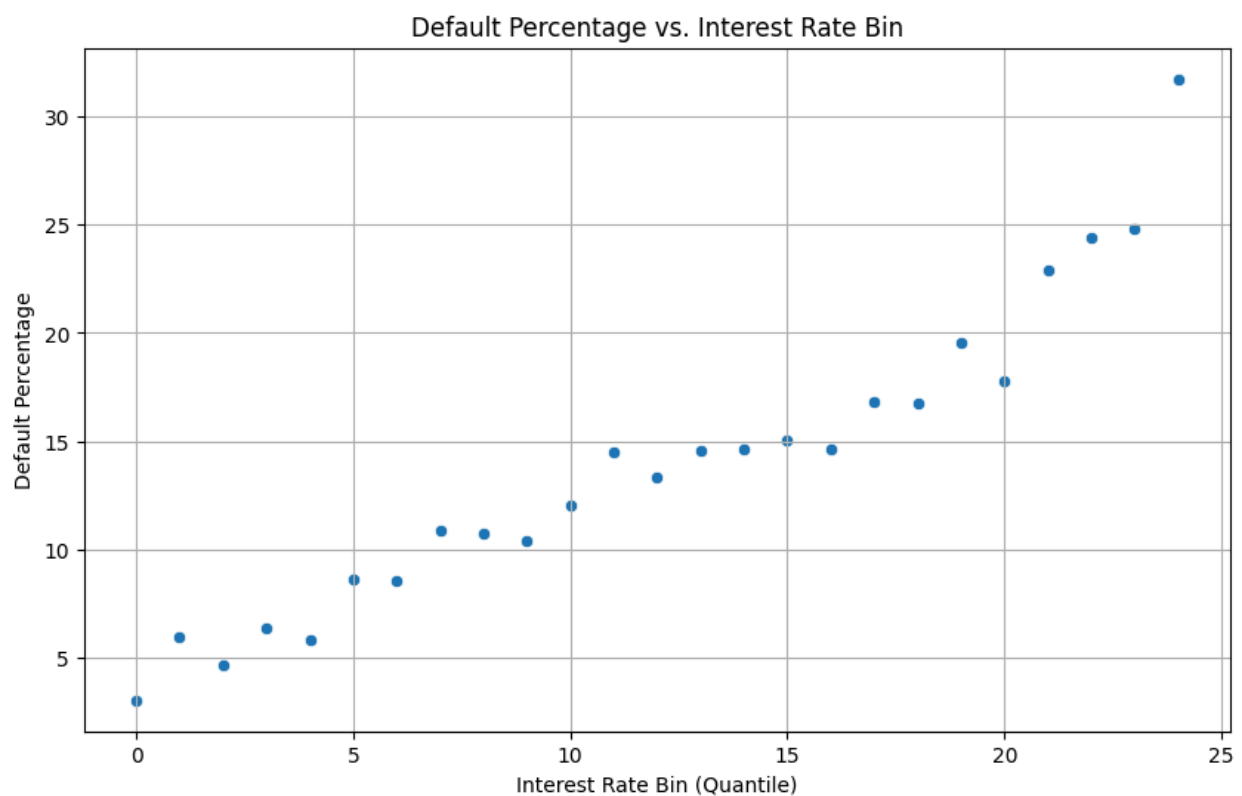
Understanding some important relationships impacting Loan Defaults

1.Default Percentage vs Interest Rate

The default percentage is calculated by the given formula:

```
int_rate_default['default_percentage'] = (int_rate_default['charged_off'] / int_rate_default['total_loans']) * 100
```

The **interest rate** is the cost of borrowing money, expressed as a percentage of the loan amount. It represents what lenders charge borrowers for using their funds over a specific period.



Analysis: Default Percentage vs. Interest Rate Bin

1) What does the relationship between interest rate and default percentage indicate?

The scatter plot of Default Percentage vs. Interest Rate Bin reveals a clear upward trend: as the interest rate increases, the default percentage rises sharply. In the lowest interest rate bins, default rates hover around 3–5%, while in the highest-rate bins, they climb beyond 30%. This suggests a strong positive correlation between the two variables.

2) Does this mean higher interest rates cause higher defaults?

Not necessarily. While the correlation is strong, causation is more complex. Borrowers who pose higher credit risks are usually assigned higher interest rates by lenders as compensation for the perceived risk. The same underlying risk factors such as poor credit history, unstable employment, or high debt-to-income ratio (DTI) also make them more prone to default. Thus, borrower riskiness is the true causal driver, influencing both interest rate assignment and default probability.

Pearson correlation coefficient: 0.96

3) How should the financial institution interpret this trend?

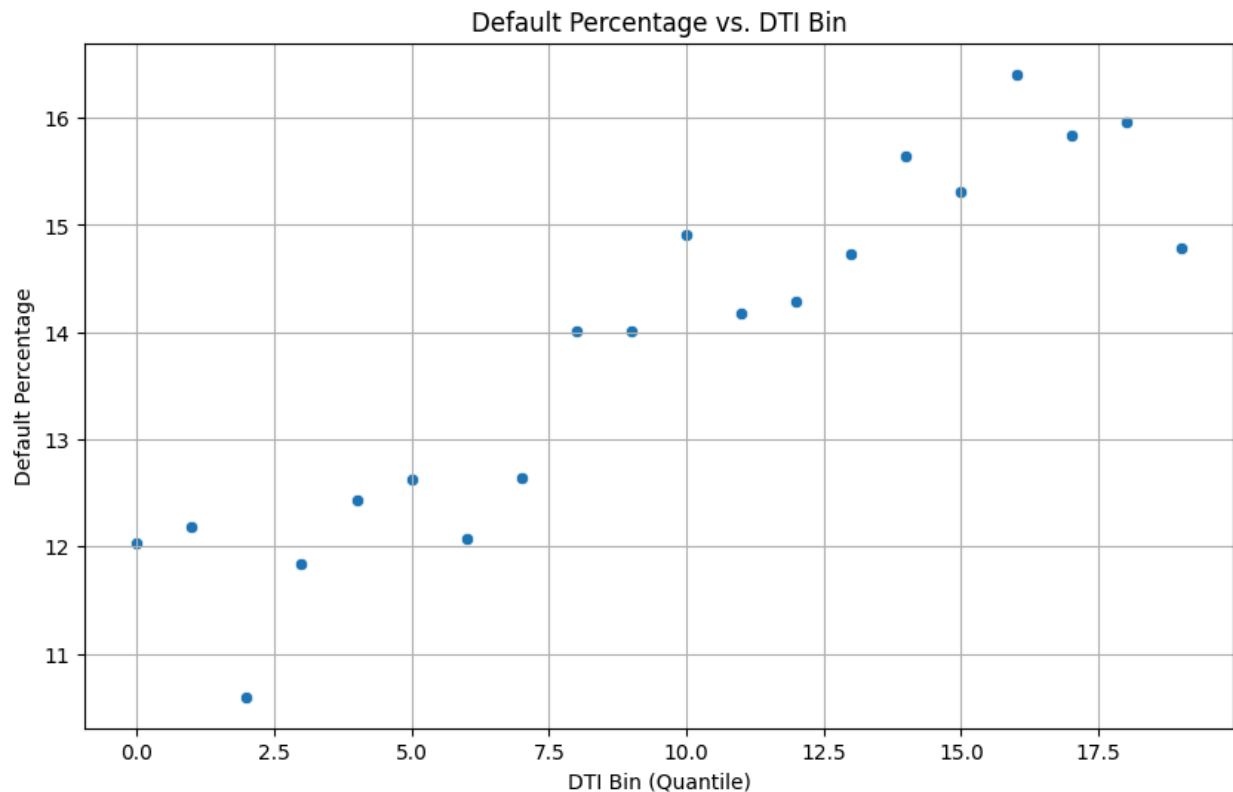
Lenders should avoid assuming that increasing interest rates directly causes borrowers to default. Instead, this pattern highlights that risk-based pricing correctly captures higher-risk segments. The focus should be on identifying and managing the underlying borrower characteristics that lead to both high interest rates and defaults.

Conclusion:

The graph demonstrates correlation, not causation. It reflects the risk segmentation inherent in lending models, where riskier borrowers receive higher rates and are also more likely to default. Recognizing this distinction helps in refining credit risk assessments, designing fair interest rate structures, and improving overall portfolio stability.

2.Default Percentage vs DTI

Debt-to-Income (DTI) is a financial metric that measures the proportion of a borrower's monthly debt payments relative to their monthly income. It is calculated by dividing total monthly debt obligations by gross monthly income and multiplying by 100 to express it as a percentage.



Analysis: Default Percentage vs. DTI Bin

1. How does DTI affect the default percentage?

The graph demonstrates a strong, positive, and non-linear relationship between a borrower's DTI and their probability of default.

- **Positive Trend:** The line on the chart trends distinctly upward from left to right. This means that as we move from the lowest DTI quantile (borrowers with the least debt relative to income) to the highest DTI quantile (the most indebted borrowers), the percentage of those borrowers who default increases substantially.

In short, higher DTI unequivocally predicts a higher chance of default, with risk escalating rapidly at the highest debt levels.

2. Is the relationship simply correlation, or does high DTI cause default?

While the graph itself illustrates a correlation, the underlying economic reality strongly suggests a causal relationship. A high DTI is not just a number on an application; it is a direct measure of financial fragility.

- **Correlation:** The data shows that high DTI and high default rates are statistically associated. We observe them happening together.

Pearson correlation coefficient: 0.92

- **Causation:** A high DTI directly causes a higher default risk by eroding a borrower's financial safety margin. Borrowers with high DTI have minimal disposable income after making their monthly debt payments. When faced with an inevitable financial shock—such as a car breakdown, medical emergency, or reduction in work hours—they lack the cash flow to absorb the blow. This forces them to choose between essential expenses and debt payments, often leading to default. The high DTI is the primary condition that enables such a shock to trigger a default.

3. How should the financial institution act on this insight?

This data provides a clear mandate for strategic action across our lending operations. We must integrate DTI as a dynamic, core component of our risk management framework.

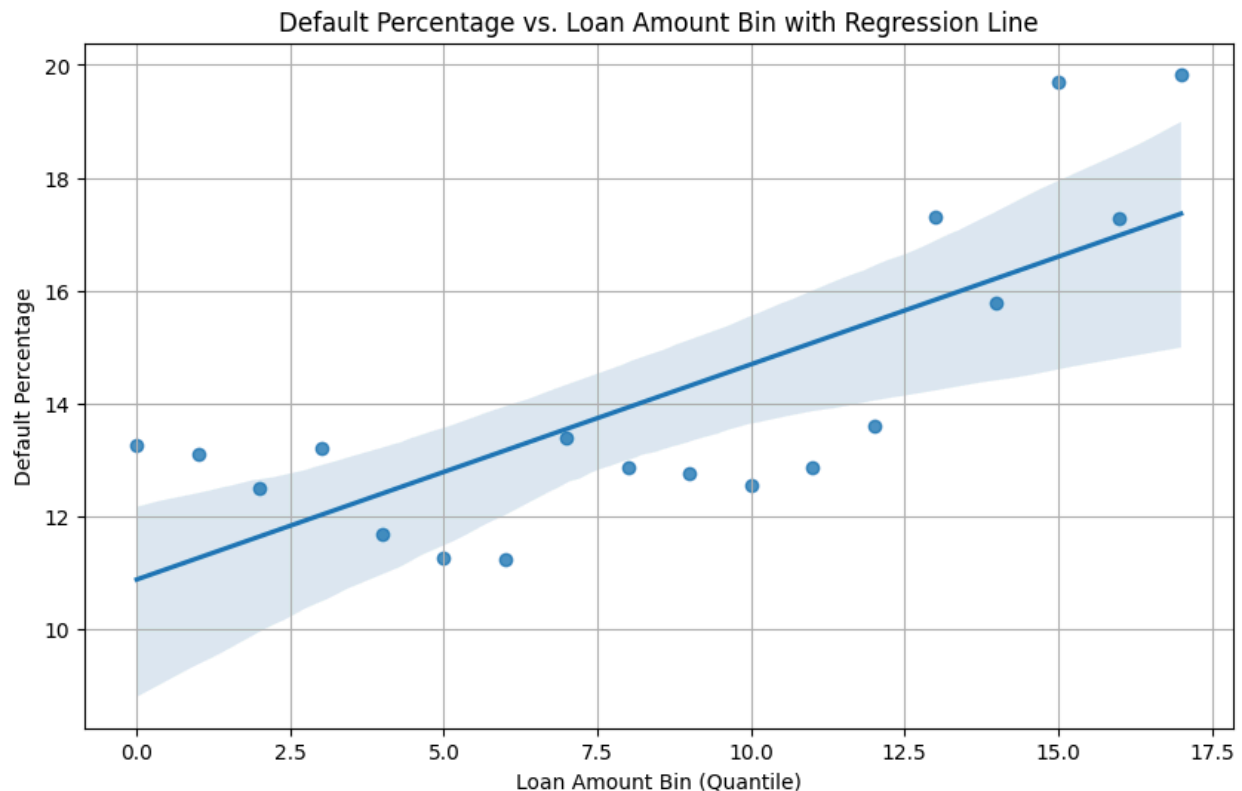
- **Implement Granular, Risk-Based Pricing:** Move beyond a binary "accept/reject" model. Develop pricing tiers where interest rates are directly calibrated to the DTI quantile. Applicants in the highest bins should be priced to reflect the significantly higher expected loss, protecting margins. Conversely, offer preferential rates to low-DTI borrowers to attract this high-quality segment.
- **Adopt Proactive Portfolio Management:** Segment the existing loan portfolio by DTI to identify at-risk borrowers before they default. For clients in the upper DTI quantiles, initiate proactive outreach. This could include offering financial health check-ups, debt consolidation solutions, or information about hardship programs. This strategy reduces future losses and builds long-term customer loyalty.
- **Establish Conservative, Data-Driven Credit Policies:** While regulators may set maximum DTI limits, internal policy should be more conservative. Default risk becomes pronounced well before the absolute maximum DTI is reached. Define risk appetite and set stricter internal DTI thresholds to limit exposure to the most volatile segments of the market.

Conclusion:

The evidence is clear and compelling: DTI is a powerful, causal predictor of borrower default. For our institution, this means that DTI must be elevated from a simple underwriting checkbox to a central pillar of strategic decision-making. By implementing risk-based pricing, proactive portfolio monitoring, and conservative internal limits, we can effectively manage risk, protect profitability, and build a more resilient and sustainable lending business.

3.Default Percentage vs loan Amount

We earlier saw how the number or percentage of loans ending up as Default loans have a positive correlation with Interest Rate on the loan taken and DTI ratio of an individual. Now we will understand the relationship between loan Amount, the actual capital lent by the institute and the loan ending up as paid or default.



1) What does the relationship between loan amount and default percentage indicate?

The analysis shows a strong positive correlation between **loan amount** and **default percentage**. Higher loan amounts are generally associated with higher default rates. Borrowers with smaller loans tend to have lower default percentages, while those with larger loans are more likely to default. This indicates that as the loan amount increases, the probability of default rises significantly.

Pearson correlation coefficient: 0.76

2) Does this mean larger loans cause higher defaults?

Not necessarily. While the correlation is strong, it does **not imply causation**. High-risk borrowers are often approved for larger loans due to factors such as higher income, longer credit history, or greater financial commitments. These same risk factors—such as unstable employment, high debt-to-income ratio (DTI), or previous delinquencies—also make them more prone to default. Thus, underlying borrower risk is the true driver influencing both loan size and default probability.

3) How should the financial institution interpret this trend?

Lenders should not assume that simply issuing larger loans causes defaults. Instead, the observed trend reflects **risk-based loan allocation**, where borrowers with higher inherent risk tend to take on larger loans and are also more likely to default. The focus should be on **identifying and managing borrower risk factors** to mitigate defaults, rather than restricting loan amounts arbitrarily.

Interpretation of Correlations

1. Strength of Relationship

- DTI and Interest Rate: $r = 0.9$. This shows an extremely strong positive correlation with defaults. Borrowers' debt burden and the interest rates they are charged are very closely associated with default probability
- Loan Amount: $r = 0.76$. This is a strong positive correlation but weaker than DTI or interest rate. Larger loans are associated with higher defaults, but the relationship is less direct and influenced by other factors

2. Underlying Implications

- Higher r (0.9) for DTI and interest rate: These are key risk indicators. Lenders can use them to more accurately identify high risk borrowers because they directly reflect borrower financial stress and risk based pricing

- Lower r (0.76) for loan amount: Loan size is still important, but defaults are not solely driven by how large the loan is. Other variables such as DTI, income, interest rate, and creditworthiness play a stronger role. Large loans may correlate with defaults, but the correlation is partially influenced by borrower risk profile

3. Practical Takeaways for Lenders

- Focus on DTI and interest rate as primary predictors of default because they carry the most predictive power
- Loan amount is a secondary predictor. While higher loans are riskier, the risk must be evaluated in conjunction with DTI and interest rate
- Recognize that high correlation does not imply causation. A large loan does not inherently cause default, just as a high DTI or interest rate does not directly cause it. All are indicators of underlying borrower risk

Summary Statement

DTI and interest rate, with $r = 0.9$, are the strongest indicators of default, reflecting the borrower's financial stress and risk based pricing. Loan amount, with $r = 0.76$, is also positively correlated with defaults but less strongly, indicating that defaults depend more on underlying risk factors than on loan size alone. This hierarchy helps lenders prioritize variables when assessing credit risk.